David Saeb, M.S.

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EDUCATION

University of Colorado Boulder — Boulder, CO

Ph.D. in Chemical Engineering — Expected May 2027
Advisor: Dr. Kayla Sprenger
Thesis Committee: Dr. Kayla Sprenger, Dr. Kimberley Bruce, Dr. Kristi Anseth, Dr. Michael Shirts, Dr. Alex Whiteley

M.S. in Chemical Engineering — May 2024

B.S. in Chemical and Biological Engineering, Music Minor — May 2023

California Institute of Technology — Pasadena, CA

Visiting Student Researcher — Feb. 2026 - Present
Advisor: Dr. Frances Arnold

AWARDS AND HONORS

Kern Lipid Conference: Early Career Investigator Award — Awarded - 2026
Lindau Nobel Laureate Meeting (Interdisciplinary) — Selected Young Scientist - 2026
AAIC Advancements: Tau Global Conference Fellowship — Awarded - 2025
AAIC Advancements: APOE & Lipid Biology Conference Fellowship — Awarded - 2025
National Science Foundation Graduate Research Fellowship (NSF-GRFP) — Awarded - 2024
Research Award - College of Engineering; *University of Colorado*, Boulder, CO — Awarded - 2023
Marilyn and Howard L. Anseth Outstanding Undergraduate Research Assistant Award — Awarded - 2023
Outstanding Undergraduate of the College; *University of Colorado*, Boulder, CO — Department Nominee - 2023
D.E. Shaw Research Undergraduate and Master's Fellowship — Awarded - 2023

PUBLICATIONS

Li, Z.-Q., Xie, P.-P., Qin, Z.-Y., Zhang, Y., **Saeb, D.**, Domecillo, M., Hanley, D., Liu, P., & Arnold, F. Biocatalytic Synthesis of Pyrrolidines via Regio- and Enantioselective C–H Amination, (Under Review at *Nature Catalysis*).

Fuchs, D., Serio, L., **Saeb, D.**, & Sprenger, K. Structural and Dynamic Insights into Small-Molecule Inhibition and Allosteric Modulation of the Human 20S Proteasome, *(Submitted).*

Dong, Z.*, Wegner, A.*, **Saeb, D.,** Cintrón Baerga, P., Kwan, M., Toney, M., Shirts, M., & Sprenger, K. Enhanced Sampling Molecular Simulations of Protein-Polymer Interactions in PEGylated Lysozyme Bioconjugates, (Under Review at *JCTC*).

Lietzke, E., **Saeb, D.,** Aldrich, E., de Menezes, R., Bruce, K., & Sprenger, K. Uncovering the Biological Mechanisms of TREM2 with Molecular Simulations: A Comprehensive Review and Perspective, *Prog. Biophys. Mol. Biol.,* 2026, DOI: 10.1016/j.pbiomolbio.2026.06.002.

Lietzke, E., **Saeb, D.**, Aldrich, E., Bruce, K.*, & Sprenger, K*. Synergistic reduction in interfacial flexibility of TREM2R47H and ApoE4 may underlie AD pathology, *Alzheimer's and Dementia,* 2025, DOI: 10.1002/alz.70120.

Saeb, D., Lietzke, E., Fuchs, D., Aldrich, E., Bruce, K., & Sprenger, K. The flexible stalk domain of sTREM2 modulates its interactions with brain-based phospholipids, *eLife*, 2025, DOI: 10.7554/eLife.102269.3.

Saeb, D., Lietzke, E., Wood, E., Aldrich, E., Fuchs, D., Chaghajerdi, A., Bruce, K., & Sprenger, K. Investigating the therapeutic potential of soluble TREM2 interactions with Transgelin-2, (In final preparation).

Saeb, D., Lietzke, E., Aldrich, E., Cintrón Baerga, P., Bruce, K & Sprenger, K. Structural differences in TREM2 alternate spliced isoforms facilitate their distinct endogenous functions, (In preparation).

PRESENTATIONS

A Lipid Membrane Embedded Model of the Full TREM2–DAP12 Signaling Complex, *Selected Early Career Investigator Oral Presentation*, Kern Lipid Conference (2026).

Molecular Modeling of TREM2 and sTREM2 Reveals Mechanisms of Microglial Signaling and Therapeutic Potential in Alzheimer's Disease, *Oral Presentation*, AIChE Annual Meeting, (Nov 2025).

Atomistic Simulations and Ligand Screening of PEG10 Aspartic Protease in ALS, *Poster Presentation*, AIChE Annual Meeting, (Nov 2025).

The flexible stalk region of sTREM2 modulates its interactions with phospholipids in the brain, *Poster Presentation,* AAIC Advancements: APOE and Lipid Biology Conference (2025).

The flexible stalk region of sTREM2 modulates its interactions with phospholipids in the brain, *Poster Presentation,* Kern Lipid Conference (2024).

Determining the molecular mechanisms of soluble TREM2 in Alzheimer's disease, *Oral Presentation,* Master's Thesis Defense (2024).

Molecular Dynamics Insights Into Soluble TREM2 Interactions with Phospholipid Activators, *Poster Presentation,* DLA Research Symposium (2023).

Molecular Dynamics Insights Into Soluble TREM2 Interactions with Phospholipid Activators, *Oral Presentation,* Senior Thesis Defense (2023).

EXPERIENCE

Visiting Student Researcher, *Arnold Lab*; *Caltech*, Pasadena, CO Feb. 2026 - Present

- Apply physics-based and physics-trained computational tools to utilize conformational dynamics in enzyme design.

Graduate Research Assistant, *Sprenger Lab*; *University of Colorado*, Boulder, CO Aug. 2023 - Present

- Utilize molecular simulations to understand protein-protein and protein-ligand interactions underlying neurodegenerative disease.
    - **Primary Research Area 1:** Investigate (s)TREM2 structural dynamics to gain insight into roles and therapeutic potential in the context of Alzheimer's disease.
    - **Primary Research Area 2:** Conduct molecular simulations of PEG10 and screen novel small molecule candidates for therapeutic potential in ALS.
- Mentor and train multiple undergraduate students in preparation for graduate school.

Advanced Teaching Assistant; *University of Colorado*, Boulder, CO Aug. 2025 - Present

- Work with faculty to coordinate course logistics, manage accommodations, design problem sets, teach recitation, and lecture for a 400+ person class.

Endoscopy Technician; *Gastroenterology of the Rockies*, Boulder, CO Feb. 2021 - Dec. 2024

- Assisted healthcare team to provide patient care for colonoscopy/endoscopy procedures.

Course Assistant; *University of Colorado*, Boulder, CO Jan. 2023 - May 2023

- Held office hours and graded homework for kinetics/biokinetics undergraduate course.

Ethics and Internal Affairs Manager; *CU Student Government*, Boulder, CO Sept. 2022 - May 2023

- Responsible for mediating conflicts within CU Student Government, advising decisions based on ethical and moral considerations, and investigating ethical shortcomings within the student government itself.

ACADEMIC SERVICE

Reviewer, GEM Workshop on Generative and Experimental Perspectives for Biomolecular Design, **NeurIPS** 2026
Reviewer, GEM Workshop on Generative and Experimental Perspectives for Biomolecular Design, **ICLR** 2026

LEADERSHIP EXPERIENCE

Graduate Student Advisory Board (*CU Boulder College of Engineering*), Boulder, CO Sept. 2024 - May 2025

- Worked to advocate for and represent the graduate student community.

Chapter President; *AMSA (American Medical Student Association)*, Boulder, CO April 2021 - May 2023

- Led a 150+ person chapter supporting pre-health students.
- Founded a chapter Volunteer Committee to plan volunteering and chapter outreach events. Successfully organized a campus 5K race to fundraise for the American Cancer Society.

Clinical Trial Lead; *BMES (Biomedical Engineering Society)*, Boulder, CO Dec. 2019 - Dec. 2020

- Responsible for organizing a 5-10 person projects team, writing clinical trial protocols, obtaining IRB approval, and coordinating with physician and faculty sponsors.

VOLUNTEER EXPERIENCE

Habitat for Humanity; Denver Health; AMSA Volunteer Committee; CU Boulder, Chemical and Biological Engineering, Peer Mentor Program Mentor; Project CURE Denver